Assignment No:01

Academic Year: 2026-27
Subject: DSA Lab
Division: IT-A
Batch: AS2

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Code:

```
#include <stdio.h>
#include <string.h>

#define MAX 100

struct Observation
{
    int id;
    char location[50];
    char timestamp[30];
    float temperature;
    float humidity;
};

struct Observation queue[MAX];
int front = -1, rear = -1;

void enqueue()
{
    struct Observation obs;

    if (rear == MAX - 1)
    {
        printf("\nQueue Overflow! Queue is full.\n");
        return;
    }

    printf("Enter Observation ID: ");
    scanf("%d", &obs.id);

    printf("Enter Location: ");
    scanf(" %49[^\n]", obs.location);

    printf("Enter Timestamp (DD-MM-YYYY HH:MM): ");
    scanf(" %29[^\n]", obs.timestamp);

    printf("Enter Temperature: ");
    scanf("%f", &obs.temperature);
```

```
    printf("Enter Humidity: ");
    scanf("%f", &obs.humidity);

    if (front == -1)
        front = 0;

    rear++;
    queue[rear] = obs;

    printf("\nObservation added successfully!\n");
}

void dequeue()
{
    if (front == -1 || front > rear)
    {
        printf("\nQueue Underflow! Queue is empty.\n");
        return;
    }

    printf("\nRemoved Observation ID: %d\n", queue[front].id);

    if (front == rear)
    {
        front = rear = -1;
    }
    else
    {
        front++;
    }
}

void peek()
{
    if (front == -1)
    {
        printf("\nQueue is empty.\n");
        return;
    }
```

```
void peek()
{
    if (front == -1)
    {
        printf("\nQueue is empty.\n");
        return;
    }

    printf("\nNext Observation to Process:\n");
    printf("ID: %d\n", queue[front].id);
    printf("Location: %s\n", queue[front].location);
    printf("Timestamp: %s\n", queue[front].timestamp);
    printf("Temperature: %.2f C\n", queue[front].temperature);
    printf("Humidity: %.2f%%\n", queue[front].humidity);
}

void display()
{
    int i;

    if (front == -1)
    {
        printf("\nQueue is empty.\n");
        return;
    }

    printf("\n--- Weather Observations ---\n");

    for (i = front; i <= rear; i++)
    {
        printf("\nObservation ID: %d\n", queue[i].id);
        printf("Location: %s\n", queue[i].location);
        printf("Timestamp: %s\n", queue[i].timestamp);
        printf("Temperature: %.2f C\n", queue[i].temperature);
        printf("Humidity: %.2f%%\n", queue[i].humidity);
    }
}
```

```c
void search()
{
    int id, i, found = 0;

    printf("Enter Observation ID to search: ");
    scanf("%d", &id);

    for (i = front; front != -1 && i <= rear; i++)
    {
        if (queue[i].id == id)
        {
            printf("\nObservation Found!\n");
            printf("ID: %d\n", queue[i].id);
            printf("Location: %s\n", queue[i].location);
            printf("Timestamp: %s\n", queue[i].timestamp);
            printf("Temperature: %.2f C\n", queue[i].temperature);
            printf("Humidity: %.2f%%\n", queue[i].humidity);

            found = 1;
            break;
        }
    }

    if (!found)
        printf("\nObservation not found.\n");
}

void count()
{
    if (front == -1)
        printf("\nTotal Observations: 0\n");
    else
        printf("\nTotal Observations: %d\n", rear - front + 1);
}
```

```c
void queueStatus()
{
    if (front == -1)
        printf("\nQueue Status: EMPTY\n");
    else if (rear == MAX - 1)
        printf("\nQueue Status: FULL (array limit reached)\n");
    else
        printf("\nQueue Status: NOT EMPTY\n");
}

int main()
{
    int choice;

    do
    {
        printf("\n===== CLIMATE HEATWAVE MONITORING SYSTEM =====\n");
        printf("1. Enqueue Observation\n");
        printf("2. Dequeue Observation\n");
        printf("3. Peek Observation\n");
        printf("4. Display Observations\n");
        printf("5. Search Observation\n");
        printf("6. Count Observations\n");
        printf("7. Queue Status\n");
        printf("8. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                enqueue();
                break;

            case 2:
                dequeue();
                break;
```

```
        {
            case 1:
                enqueue();
                break;

            case 2:
                dequeue();
                break;

            case 3:
                peek();
                break;

            case 4:
                display();
                break;

            case 5:
                search();
                break;

            case 6:
                count();
                break;

            case 7:
                queueStatus();
                break;

            case 8:
                printf("\nExiting Program...\n");
                break;

            default:
                printf("\nInvalid Choice! Try Again.\n");
        }

    } while (choice != 8);

    return 0;
}
```

Output:

```
PS C:\Users\HP\Desktop> gcc Suhani.c -o test.exe
PS C:\Users\HP\Desktop> .\test.exe

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 1
Enter Observation ID: 101
Enter Location: Mumbai
Enter Timestamp (DD-MM-YYYY HH:MM): 12-10-2026 14:30
Enter Temperature: 38.5
Enter Humidity: 72

Observation added successfully!

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 1
Enter Observation ID: 102
Enter Location: Pune
Enter Timestamp (DD-MM-YYYY HH:MM): 12-10-2026 15:00
Enter Temperature: 40.2
Enter Humidity: 65

Observation added successfully!

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
```

```
===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 1
Enter Observation ID: 103
Enter Location: Nashik
Enter Timestamp (DD-MM-YYYY HH:MM): 12-10-2026 15:30
Enter Temperature: 36.8
Enter Humidity: 80

Observation added successfully!

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 4

--- Weather Observations ---

Observation ID: 101
Location: Mumbai
Timestamp: 12-10-2026 14:30
Temperature: 38.50 C
Humidity: 72.00%

Observation ID: 102
Location: Pune
Timestamp: 12-10-2026 15:00
```

```
Observation ID: 102
Location: Pune
Timestamp: 12-10-2026 15:00
Temperature: 40.20 C
Humidity: 65.00%

Observation ID: 103
Location: Nashik
Timestamp: 12-10-2026 15:30
Temperature: 36.80 C
Humidity: 80.00%

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 3

Next Observation to Process:
ID: 101
Location: Mumbai
Timestamp: 12-10-2026 14:30
Temperature: 38.50 C
Humidity: 72.00%

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 2
```

```
Enter your choice: 2

Removed Observation ID: 101

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 4

--- Weather Observations ---

Observation ID: 102
Location: Pune
Timestamp: 12-10-2026 15:00
Temperature: 40.20 C
Humidity: 65.00%

Observation ID: 103
Location: Nashik
Timestamp: 12-10-2026 15:30
Temperature: 36.80 C
Humidity: 80.00%

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 5
Enter Observation ID to search: 103
```

```
Observation Found!
ID: 103
Location: Nashik
Timestamp: 12-10-2026 15:30
Temperature: 36.80 C
Humidity: 80.00%

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 6

Total Observations: 2

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 7

Queue Status: NOT EMPTY

===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
```

```
===== CLIMATE HEATWAVE MONITORING SYSTEM =====
1. Enqueue Observation
2. Dequeue Observation
3. Peek Observation
4. Display Observations
5. Search Observation
6. Count Observations
7. Queue Status
8. Exit
Enter your choice: 8

Exiting Program...
PS C:\Users\HP\Desktop>
```